Disease - transmission and control

1 Classify the following diseases under the headings 'Transmissible' and 'Non-transmissible'.
lung cancer, whooping cough, rickets, arthritis, appendicitis, food poisoning, tuberculosis, measles, diabetes, anaemia, syphilis, influenza, AIDS, coronary heart disease, haemophilia

2 (a) In droplet infection (i) where do the droplets come from, (ii) what infective agents might they contain?
(b) Give two examples of diseases normally spread by droplets.

3 (a) Give two examples of diseases which can be spread by contaminated drinking water.
(b) How can the spread of such diseases be prevented?

4 Suggest three ways in which food might become contaminated by harmful bacteria.

5 Which one of the following is least important in preventing the contamination of food?
(a) Keeping food away from flies.
(b) Washing the hands before preparing food.
(c) Not coughing or sneezing over food.
(d) Not smoking while preparing food.

6 The following events explain an outbreak of food poisoning in a factory canteen. Put the events in the appropriate order.
(a) Pathogenic bacteria in intestine.
(b) Unwashed hands prepare food.
(c) Kitchen worker has intestinal disease.
(d) Outbreak of food poisoning.
(e) Pathogenic bacteria in faeces.
(f) Pathogenic bacteria get into food.
(g) Hands not washed after defecation.
(h) Faeces expelled in lavatory.
(i) Food ingested by healthy people.

7 Which one of the following is the most appropriate description of a contagious disease?
A contagious disease is one which is spread by
(a) contaminated food,
(b) contaminated drinking water,
(c) contact with an infected person,
(d) droplets from an infected person.

8 In the air passages, a layer of ….. (A) ….. traps bacteria. This ….. (B) …..is carried away from the lings by the action of …..(C) …..cells which line the air passages.

9 The ….. (A) ….. layer of the skin is a barrier to the entry of bacteria. Also the …..(B) ….. glands and ….. (C) ….. glands produce bacteriocidal substances. If bacteria get through damaged skin they are attacked by ….. (D) …..and ….. (E) …..in the blood.

10 What is the purpose of the gauze mask worn by surgeons during an operation?

11 What sources of water are (a) most likely, (b) least likely to contain pathogenic bacteria?

12 Which one of the following is least likely to prevent contamination of food by houseflies?
(a) A well-fitting dustbin lid.
(b) Swatting flies.
(c) Covering food with cling-film.
(d) Efficient sewage disposal.

13 Complete the following sentences by selecting the appropriate words from the list below.
In many cases, when you catch an infectious disease, your blood produces ….. (A) ….. against the infective organism. These ….. (B) ….. remain in the blood or can be rapidly made so that you are ….. (C) ….. to further attacks of the disease.
You can acquire ….. (D) …..by receiving injections of a …..(E) ….. form of the pathogen or its inactivated …..(F) ….. The injected substance is called a ….. (G) …..

immune, immunity, pathogens, disease, vaccine, antibodies, toxin, harmless, antigens

14 Which of the following diseases can be prevented by inoculation?

rubella, syphilis, tuberculosis, polio, gonorrhoea, common cold, measles, haemophilia, mumps, tetanus, AIDS

15 Match each disease with its correct type of vaccine.

(a) tuberculosis	(i) harmless forms of virus
(b) whooping cough	(ii) inactivated toxin
(c) diphtheria	(iii) killed bacteria
(d) polio	(iv) harmless form of bacteria.

16 Antiseptics and disinfectants both kill ….. (A) …..but only ….. (B) ….. are safe to use on the skin (if sufficiently dilute).

17 Complete the following sentence, selecting the most appropriate words from the list below.

An ideal …..(A) ….. against an infectious disease would be one which destroyed the ….. (B) …...but did not harm the …..(C) …..
pathogen, host, drug

18 A person with a cough takes a patent cough syrup. In three days, the cough is better. Does this mean that the syrup has cured the cough? Justify your answer.

19 Which one of the following is least likely to give rise to food poisoning?
(a) Cooking and eating a partially defrosted turkey.
(b) Eating a cooked chicken leg straight from the refrigerator.
(c) Preparing a fresh chicken for the oven and immediately making a lettuce and tomato salad.
(d) putting an uncooked chicken on the same plate as some cold ham.

20 Which of the following are not ways in which gonorrhoea and syphilis can be transmitted?
(a) Using a towel that has been used by an infected person.
(b) Touching an infected person.
(c) Kissing an infected person.

(d) Having sexual intercourse with an infected person.